

Qingxiu Liu

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Education

The Chinese University of Hong Kong (CUHK)

Aug 2023 – Jul 2027 (expected) Hong Kong

- Ph.D. Student in Computer Science

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS)

Sep 2020 – Jun 2023 Beijing

- M.E. in Electronic and Information Engineering, GPA 3.76/4.0

Beijing University of Posts and Telecommunications (BUPT)

Sep 2016 – Jun 2020 Beijing

- B.E. in Information Security, GPA 3.56/4.0

Research Experiences

Memory Optimization in MoE inference

Project Principal.

Sep 2025 – Apr 2026 SCITIX AI (Mentor: Dr. Yongchao He)

- Formalize expert paging, a new execution model that decouples the logical identity of experts from their physical GPU residency. Reclaim GPU memory for KV cache by treating expert parameters as transient and streamed resources.
- Design and implement FluxMoE, an MoE inference system with three key designs: PagedTensor for virtualizing tensor addresses, a bandwidth-balanced memory hierarchy that co-optimizes loading rates across compressed GPU memory and host DRAM, and a budget-aware residency planner that dynamically selects expert residency under KV pressure.

In-Switch File-System Metadata Cache

Project Principal.

Sep 2023 – Jul 2025 CUHK (Supervisor: Professor Patrick P. C. Lee)

- Design and implement Fletch, an in-switch file-system metadata caching system to provide high throughput and low latencies for distributed file-system metadata services.
- Address path dependencies in in-switch caching. Design a path-aware cache admission and eviction policy for good cache efficiency, an in-switch read-write locking for efficient concurrent reads and writes, and a token-based hash collision resolution method to ensure correct metadata retrieval.

Privacy-Preserving Neural Network Inference

Project Principal.

Dec 2019 – Jun 2023 ICT, CAS (Supervisor: Professor Qun Huang)

- Design and implement PP-Stream, a two-party privacy-preserving neural network inference service based on homomorphic encryption and advanced obfuscation mechanism.
- Apply stream processing architecture and two-level resource allocation to improve performance of PP-Stream. In the first level, formalize the resource allocation problem as ILP. In the second level, propose an algorithm to achieve load balancing of each worker.
- Prototype and deploy several mainstream neural networks on PP-Stream.

Two Projects on Intel SGX

Project Principal.

📅 May 2018 – Dec 2019

📍 IIIS, THU (Supervisor: Professor Wenfei Wu)

- Design and build a system called SecNF based on Intel SGX to protect user data (i.e., data to be processed within NFs) in the outsourcing service scenario.
- Design and build a trusted third-party DNS service T^2DNS based on Intel SGX to protect user privacy (i.e., association between a client IP and the queried names).

Publications

- [In Submission] FluxMoE: Decoupling Expert Residency for High-Performance MoE Serving.
Qingxiu Liu, Yongchao He, Runhan Jiang, Zion Wang, Bohan Zhao, Mi Zhang, and Patrick P. C. Lee.
- [IEEE TON] Fletch: File-System Metadata Caching in Programmable Switches. (CCF-A)
Qingxiu Liu, Jiazhen Cai, Siyuan Sheng, Yuhui Chen, Lu Tang, Zhirong Shen, and Patrick P. C. Lee.
- [IEEE ICDE 2024] PP-Stream: Toward High-Performance Privacy-Preserving Neural Network Inference via Distributed Stream Processing. (CCF-A)
Qingxiu Liu, Qun Huang, Xiang Chen, Sa Wang, Wenhao Wang, Shujie Han, and Patrick P. C. Lee.
- [IEEE ICCCN 2020] T^2DNS : A Third-Party DNS Service with Privacy Preservation and Trustworthiness. (CCF-C)
Qingxiu Liu, Wenfei Wu, Qingsong Liu, and Qun Huang.

Academic Activities

- Services
 - IEEE Transactions on Networking (TON) reviewer
 - SIGOPS ATC'26 AEC member
- Teaching Assistant in CUHK
 - CSCI4180 Introduction to Cloud Computing and Storage (Instructor: Prof. Patrick P. C. Lee): Fall 2023, Fall 2024, Fall 2025
 - AIST3020 Introduction to Computer Systems (Instructor: Prof. Patrick P. C. Lee): Spring 2024, Spring 2025

Awards

- Postgraduate Studentship in CUHK, 2023-2027
- 2nd Prize Scholarship in BUPT, 2018 and 2019
- 1st Prize in Beijing Contest District in China Undergraduate Mathematical Contest in Modeling, 2018